Rahul Tammalla

Postdam, New york | tammalr@clarkson.edu | +1 (315) 603 7792 | Rahul.portfolio | Rahul.linkedin
Rahul.github

Experience

Research assistant, Clarkson University, – Potsdam, United States. Jan 2025 – Present

- Analyzed **20M+** emergency medical records from the **NEMSIS** dataset to identify factors causing on-scene time more than **10 mins** in stroke-related 911 calls using **statistical** and **machine learning** models.
- Cleaned, transformed, and validated **20M+** records; handled **1M+** missing entries to ensure data quality across datasets.
- Engineered **70+** features and performed **hypothesis testing** to identify key predictors affecting emergency response time.
- Built and evaluated **Logistic Regression**, **Random Forest**, and **ANOVA models**; used **PostgreSQ**L for efficient querying and data management and **Tableau** for data visualization.

Associate Engineer, Cognizant Technology Solution, – Hyderabad, India. Oct 2021 – Aug 2024

- Optimized and engineered a high performance cloud-based Appian application for **Gilead Life Sciences**, a leading pharmaceutical company, processing **1M+** transactions to enhance reliability and user experience.
- Designed interactive dashboards for real-time insights, improving decision-making and visibility into key metrics.
- Improved Database and **Store procedure** performance, increasing execution speed and response time by **30%**, and ensured data integrity through daily system file analysis.
- Performed a daily in-depth data analysis on **500+** daily system records (transactions, inventory, on-hand, delivery) to ensure accuracy and seamless operations.

Education

Clarkson University, Masters in Applied Data Science. (GPA : 3.889) Aug 2024 – Dec 2025

- **Coursework:** Probability Statistics, Information visualization, Database modeling, Machine Learning, Data Mining, Data warehousing.

Amrita Vishwa Vidyapeetham, Bachelors in Computer Science Engineering. May 2017 – June 2021

Projects

AI Medical Chat Bot[Python, Neo4j, Cypher Queries, RAG, GPT, spaCy] 2025

- Developed a **hybrid chatbot** that integrates **Neo4j Knowledge Graphs** with **GPT** using **Retrieval-Augmented Generation (RAG)** to ensure **0% hallucination** for structured queries while handling general medical queries.

News Scraper[Transformers, Pandas, NewsAPI, DistilBERT, T5-small, Streamlit] 2025

- Developed app that scrapes top US news using NewsAPI, performs **sentiment analysis** using **DistilBERT**, and **summarizes articles** using **T5-small LLM model**, updated the news every 24 hours.

AI Research Paper Summarization[pdfplumber, scikit-learn, Hugging Face Transformers (T5), PyTorch,Git, Streamlit] 2025

- Developed application that **summarizes academic research papers** by extracting text from PDFs using **pdfplumber** and generating **extractive summary** using **TextRank**, producing human-like **abstractive summary** with **Hugging Face T5 LLM model**, and evaluating summary quality using **ROUGE-1** and **ROUGE-2** metrics.

Snow Depth Forecasting[ARIMA, AutoARIMA, HoltWinters] 2025

- Designed and implemented time series forecasting **Machine learning** models to predict snow depth in Tupper Lake for 2026,and **optimized** accuracy by **model evaluation**.

US Accident Insights[dplyr, lubridate, ggplot2, RShiny] 2024

- Developed the "US Accident Insights" **Shiny app**, analyzing **40K+** records of accidents across United States and visualizing traffic accident data using **R libraries** with **Geo-spatial** visualizations.

Personalized Course Recommendation Database System [SQL] 2024

- Developed a dynamic course recommendation system with **SQL triggers** and **procedures** to optimize database for exam schedules, instructor workloads, and student performance.

Skills

Programming & Development: Python, SQL, R, Java, MATLAB, Agile, Scrum, OOP , HTML & CSS , MS tools.
Libraries & Frameworks: NumPy, Pandas, SciPy, scikit-learn, PySpark, TensorFlow, Keras, OpenCV, Matplotlib, Seaborn, AWS, PyTorch, Azure.
Data Engineering & Visualization: ETL, CI/CD, DevOps, R Studio, Tableau, Power BI.
Databases & Cloud: PostgreSQL, MongoDB, Snowflake, Hadoop, Hive, Streamlit, Neo4j.
Machine Learning : Regression, Classification, Clustering, Random Forests, XGBoost, SVM, Statistical Modeling, Ensemble Learning, CNN, Neural Networks, Time series.
NLP : Hugging Face Transformers-(T5,DistilBERT), spaCy, RAG(Retrieval-Augmented Generation).

Certifications

Advanced NLP with Python for Machine Learning - [Linkedin learning] March 2025
Hands-On PyTorch Machine Learning - [Linkedin learning] March 2025